

Chemistry  
Builds  
Better  
Thinkers

Small  
Steps  
Big  
Dreams :)

NEET | JEE  
Foundation

|                 |                  |                  |                  |
|-----------------|------------------|------------------|------------------|
| 1<br>H<br>1.008 | 6<br>C<br>12.011 | 7<br>N<br>14.007 | 8<br>O<br>15.999 |
|-----------------|------------------|------------------|------------------|

Dr. HOMI BHABHA Exam

Practice Questions Set

Class: 9<sup>th</sup>

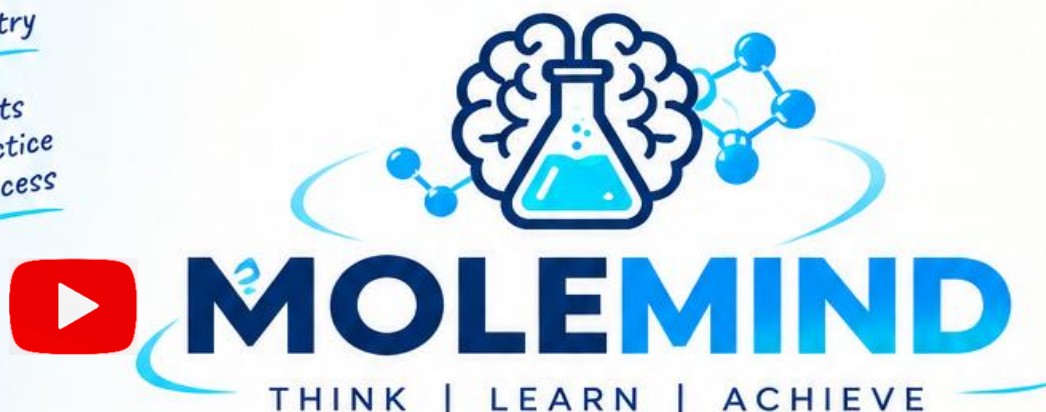
 Isomerism

 Electronic Effects

 Organic Chemistry

 Concepts  
→ Practice  
→ Success

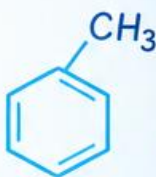
  
Practice  
Learn  
Improve  
Excel  
★



## Chapters:

- ✚ Matter, States of Matter & Changes
- ✚ Physical & Chemical Changes
- ✚ Elements, Compounds, Mixtures & Separation Techniques
- ✚ Atoms, Molecules & Structure of Atom
- ✚ Acids, Bases & Indicators

Organic  
Chemistry  
is not just a subject...  
it's a way of thinking  
❤



Better  
Concepts  
Higher  
Scores

Good  
Things  
Take  
Time ❤



From Confusion to Clarity

Dr. HOMI BHABHA Exam Class 9<sup>th</sup>

— MOLEMIND —



# Practice Question Set

## Matter, States of Matter & Changes

**Q1.** A substance occupies a definite volume but takes the shape of its container. It is most likely:

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Q2.** Which property of particles best explains why gases can be compressed easily?

- A) They have high mass
- B) They have very strong attractive forces
- C) There are large spaces between particles
- D) Their particles do not move

**Q3.** When a perfume bottle is opened in one corner of a room, its smell is detected throughout the room after some time. This phenomenon is primarily due to:

- A) Osmosis
- B) Diffusion
- C) Sedimentation
- D) Condensation

**Q4.** Which change involves absorption of heat from the surroundings?

- A) Freezing of water
- B) Condensation of steam
- C) Melting of ice
- D) Deposition of vapour

**Q5.** A student observes that ice at  $0^{\circ}\text{C}$  and water at  $0^{\circ}\text{C}$  coexist in a container. During melting, the temperature remains nearly constant because the supplied heat is mainly used for:

- A) Increasing the mass of water
- B) Increasing the kinetic energy of particles
- C) Overcoming intermolecular attractions
- D) Increasing atmospheric pressure

**Q6.** Which pair represents **two different physical states of the same substance**?

- A) Oxygen and ozone
- B) Water and steam
- C) Carbon and carbon dioxide
- D) Sulphur and sulphur dioxide

**Q7.** Which observation provides the strongest evidence that matter is made up of moving particles?

- A) A stone has a fixed shape
- B) Sugar dissolves and spreads uniformly in water
- C) Water has a fixed volume
- D) Metals are shiny

**MOLEMIND**  
THINK | LEARN | ACHIEVE

From Confusion to Clarity

— MOLEMIND —

Q8. A gas is cooled considerably. Which change is most likely?

- A) Particle motion increases
- B) Interparticle distance generally decreases
- C) Particle mass decreases
- D) Number of particles decreases

Q9. Which process is **not** a change of state?

- A) Melting
- B) Sublimation
- C) Burning
- D) Condensation

Q10. Naphthalene balls gradually disappear when left in a cupboard without forming a liquid. This is:

- A) Evaporation
- B) Condensation
- C) Sublimation
- D) Freezing

### Conceptual Traps:

Q11. Two identical amounts of water are placed in two vessels. Vessel A has a wide surface and vessel B has a narrow surface. Under otherwise identical conditions, water evaporates faster from:

- A) A
- B) B
- C) Both at the same rate
- D) Cannot be predicted

Q12. Evaporation causes cooling because:

- A) All particles lose energy
- B) Higher-energy particles escape from the surface
- C) Lower-energy particles escape first
- D) The liquid gains heat from the surroundings

Q13. Which condition would generally **decrease** the rate of evaporation?

- A) Increasing temperature
- B) Increasing surface area
- C) Increasing humidity
- D) Increasing wind speed

Q14. Steam at  $100^{\circ}\text{C}$  can cause more severe burns than water at  $100^{\circ}\text{C}$  mainly because steam:

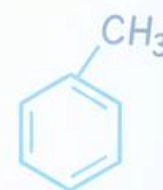
- A) Has no mass
- B) Contains latent heat that is released during condensation
- C) Has a lower temperature
- D) Is chemically different from water

Q15. A wet cloth dries faster on a windy day because wind:

From Confusion to Clarity

|             |             |             |
|-------------|-------------|-------------|
|             | 1           | 2           |
|             | H<br>1.008  |             |
| 6           | 7           | 8           |
| C<br>12.011 | N<br>14.007 | O<br>15.999 |

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

Good  
Things  
Take  
Time

- Chemistry Builds Better Thinkers
- Small Steps Big Dreams :)
- NEET | JEE Foundation
- |        |        |        |
|--------|--------|--------|
| 6      | 7      | 8      |
| C      | N      | O      |
| 12.011 | 14.007 | 15.999 |
- A) Decreases the kinetic energy of water molecules  
B) Removes water vapour near the cloth  
C) Decreases the surface area  
D) Converts water directly into ice

Q16. Which statement about boiling is correct?

- Atomizerism
- A) It occurs only at the surface  
B) It can occur throughout the liquid  
C) It occurs at every temperature  
D) It does not involve energy absorption

Q17. At high altitude, water generally boils at a temperature:

- Electronic Effects
- A) Higher than  $100^{\circ}\text{C}$   
B) Exactly  $100^{\circ}\text{C}$   
C) Lower than  $100^{\circ}\text{C}$   
D) Always  $0^{\circ}\text{C}$

Q18. If atmospheric pressure increases, the boiling point of a liquid generally:

- Organic Chemistry Concepts
- A) Decreases  
B) Increases  
C) Becomes zero  
D) Remains unchanged

Q19. Which phenomenon demonstrates that particles possess kinetic energy even when a substance appears stationary?

- Practice Success
- A) Diffusion  
B) Filtration  
C) Sedimentation  
D) Decantation

Q20. A solid changes directly into a gas. The reverse process is:

- Better Concepts Higher Scores :)
- A) Evaporation  
B) Condensation  
C) Deposition  
D) Fusion

Organic Chemistry is not just a subject it's a way of thinking

### Physical & Chemical Changes

Q21. Which is a chemical change?

- Good Things Take Time
- A) Melting wax  
B) Cutting paper  
C) Rusting iron  
D) Dissolving sugar

Q22. Which observation provides the strongest evidence of a chemical reaction?

From Confusion to Clarity

- A) Change in shape  
B) Change in size  
C) Formation of a new substance  
D) Change in physical state

**Q23.** Burning magnesium ribbon produces a white powder. The best conclusion is:

- A) Only a physical change occurred  
B) Magnesium changed into a new substance  
C) Magnesium only changed its shape  
D) No reaction occurred

**Q24.** Which change is reversible under ordinary conditions?

- A) Burning wood  
B) Rusting iron  
C) Melting ice  
D) Cooking food

**Q25.** A student says, "Dissolving salt in water is definitely a chemical change because salt disappears." Which is the best response?

- A) Correct, because disappearance always means reaction  
B) Incorrect, because salt can be recovered by evaporation  
C) Correct, because water reacts with salt  
D) Incorrect, because salt is destroyed

**Q26.** Which combination contains **only physical changes**?

- A) Melting ice, cutting paper, dissolving sugar  
B) Burning coal, rusting iron, cooking food  
C) Curdling milk, burning paper, melting wax  
D) Digestion, respiration, freezing water

**Q27.** Formation of curd from milk is considered a chemical change mainly because:

- A) Its shape changes  
B) A new substance with different properties forms  
C) Its temperature changes  
D) It becomes thicker only

**Q28.** Crystallisation of copper sulphate from its solution is generally classified as:

- A) Chemical change  
B) Physical change  
C) Nuclear change  
D) Combustion

**Q29.** Which process involves both physical and chemical changes?

- A) Burning a candle  
B) Melting ice  
C) Breaking glass  
D) Freezing water

**Q30.** When a candle burns, the wax near the flame first melts and then the vapour burns. Therefore:

- A) Entire process is physical
- B) Entire process is chemical
- C) Both physical and chemical changes occur
- D) No change occurs

### Scientific Reasoning

**Q31.** A student heats a liquid and observes bubbles throughout the liquid. Which observation would best distinguish boiling from evaporation?

- A) It occurs at the surface
- B) It occurs throughout the liquid
- C) It causes cooling
- D) It involves liquid particles

**Q32.** Four samples are given:

- P: Ice
- Q: Water
- R: Steam
- S: Liquid nitrogen

Which pair contains substances in the **same physical state**?

- A) P and Q
- B) Q and R
- C) P and S
- D) R and S

**Q33.** A sealed container contains water and water vapour at equilibrium. If some vapour condenses, then eventually:

- A) No evaporation can occur
- B) Evaporation can continue until equilibrium is restored
- C) All water becomes vapour
- D) The container becomes empty

**Q34.** Which statement is **incorrect**?

- A) Gases exert pressure
- B) Liquids have definite volume
- C) Solids have no intermolecular forces
- D) Gas particles are in continuous motion

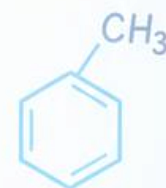
**Q35.** A liquid evaporates faster when its surface area is increased. This suggests that evaporation:

- A) Depends on the number of particles at the surface
- B) Depends only on mass
- C) Occurs only at the bottom
- D) Is independent of temperature

From Confusion to Clarity

|                  |                  |                  |
|------------------|------------------|------------------|
| 6<br>C<br>12.011 | 7<br>N<br>14.007 | 8<br>O<br>15.999 |
| 1<br>H<br>1.008  |                  |                  |

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores

Good  
Things  
Take  
Time



## Olympiad-Level Questions

**Q36.** A sealed bottle is filled completely with water and kept in a freezer. The bottle may crack because:

- A) Water contracts on freezing
- B) Water expands on freezing
- C) Ice has no volume
- D) Pressure inside becomes zero

**Q37.** Which statement best explains why a wooden block can float on water while an iron nail sinks?

- A) Wood contains no particles
- B) Density and buoyancy are involved
- C) Iron has no mass
- D) Water attracts wood chemically

**Q38.** A drop of ink is added to two beakers containing equal amounts of water. One is at  $20^{\circ}\text{C}$  and the other at  $60^{\circ}\text{C}$ . The ink spreads faster in the hotter water because:

- A) Water molecules become heavier
- B) Molecular motion increases with temperature
- C) Ink becomes a gas
- D) Water stops attracting ink

**Q39.** A student claims: "When water boils, the molecules themselves become hotter and therefore turn into steam." Which correction is most appropriate?

- A) Molecules disappear
- B) Liquid water changes state as energy enables particles to overcome intermolecular attraction
- C) Water changes into a new chemical substance
- D) Steam contains different atoms

**Q40.** Which sequence represents increasing particle freedom of movement?

- A) Gas  $\rightarrow$  Liquid  $\rightarrow$  Solid
- B) Solid  $\rightarrow$  Liquid  $\rightarrow$  Gas
- C) Liquid  $\rightarrow$  Solid  $\rightarrow$  Gas
- D) Solid  $\rightarrow$  Gas  $\rightarrow$  Liquid

**Q41.** A wet hand feels colder when exposed to moving air because:

- A) Air freezes the water
- B) Evaporation is enhanced and removes heat
- C) Water gains heat
- D) Air increases humidity

**Q42.** Which situation is most directly explained by **diffusion**?

- A) A stone sinking in water
- B) Smell of incense spreading through a room
- C) Salt settling at the bottom
- D) Water freezing

**Q43.** If the kinetic energy of particles in a substance increases sufficiently, the substance may:

A) Change its state  
B) Lose all particles  
C) Become massless  
D) Stop occupying space

**Q44.** A substance changes from solid  $\rightarrow$  liquid  $\rightarrow$  gas. Which statement is correct?

- A) Interparticle separation generally increases  
B) Particle mass continuously decreases  
C) Chemical identity must change at every stage  
D) Number of atoms necessarily decreases

**Q45.** Which change would require **maximum energy input** under suitable conditions?

- A) Freezing  
B) Condensation  
C) Melting  
D) Deposition

## Challenge Round

**Q46.** A student places equal quantities of water in four conditions:

- I. Wide open vessel
- II. Narrow open vessel
- III. Wide vessel with a fan
- IV. Wide vessel in a humid room

Which is expected to evaporate fastest?

- A) I  
B) II  
C) III  
D) IV

**Q47.** A substance has a fixed volume but no fixed shape. Its particles are:

- A) Completely fixed in position  
B) Close together but able to move past one another  
C) Extremely far apart with negligible attraction  
D) Completely motionless

**Q48.** Which pair represents **opposite changes of state**?

- A) Melting and freezing  
B) Evaporation and sublimation  
C) Boiling and melting  
D) Condensation and freezing

**Q49.** A liquid is cooled below its freezing point but remains liquid temporarily. This phenomenon is called:

- A) Superheating  
B) Supercooling  
C) Sublimation  
D) Deposition

Small  
Steps  
Big  
Dreams :)

NEET | JEE  
Foundation

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

Good Things Take Time ♡

## From Confusion to Clarity

— MOLEMIND —

NEET

JEE



FOUNDATION

**Q50.** Three statements are given:

**P:** Evaporation can occur below the boiling point.

**Q:** Boiling occurs only at the surface of a liquid.

**R:** Increasing wind speed generally increases evaporation.

Which option is correct?

- A) P only  
B) Q only  
C) P and R  
D) Q and R

## Elements, Compounds, Mixtures & Separation Techniques

**Q51.** Which of the following is a **pure substance**?

- A) Air  
B) Distilled water  
C) Brass  
D) Soil

**Q52.** Which statement correctly distinguishes a compound from a mixture?

- A) A compound always contains two or more elements physically mixed  
B) Components of a compound can always be separated by physical methods  
C) A compound has properties different from those of its constituent elements  
D) A mixture always has a fixed composition

**Q53.** Which of the following is a **homogeneous mixture**?

- A) Sand + water  
B) Oil + water  
C) Salt solution  
D) Soil

**Q54.** Which is an example of a **heterogeneous mixture**?

- A) Air  
B) Brass  
C) Sugar solution  
D) Granite

**Q55.** A sample contains only hydrogen and oxygen atoms in a fixed ratio of 2:1. The sample is most likely:

- A) A mixture of hydrogen and oxygen  
B) Water  
C) Oxygen dissolved in hydrogen  
D) Hydrogen peroxide

**Q56.** Which statement about a mixture is correct?

- A) Its components always combine chemically  
B) Its composition is always fixed

## From Confusion to Clarity

— MOLEMIND —

D) It always has a new chemical formula

**Q57.** Which pair contains **only compounds**?

- A) Water and carbon dioxide  
B) Oxygen and nitrogen  
C) Brass and steel  
D) Air and water

**Q58.** Which pair contains **only mixtures**?

- A) Air and brass  
B) Water and carbon dioxide  
C) Oxygen and nitrogen  
D) Sodium chloride and air

**Q59.** A substance cannot be broken down into simpler substances by ordinary chemical methods. It is classified as:

- A) Mixture  
B) Compound  
C) Element  
D) Solution

**Q60.** Which of the following is **not** a characteristic of a pure compound?

- A) Fixed composition  
B) Definite properties  
C) Components retain all their original properties  
D) Chemical combination of elements

## Conceptual Traps

**Q61.** A mixture contains iron filings, sand and common salt. Which sequence would be most appropriate to separate all three?

- A) Filtration → magnet → evaporation  
B) Magnet → dissolve in water → filtration → evaporation  
C) Evaporation → magnet → filtration  
D) Filtration → evaporation → magnet

**Q62.** Why is a magnet used first in the separation of iron, sand and salt?

- A) Salt is magnetic  
B) Iron can be selectively removed without affecting sand and salt  
C) Sand becomes magnetic in water  
D) Salt reacts with the magnet

**Q63.** A mixture contains ammonium chloride and sand. The most suitable method for separating ammonium chloride is:

- A) Filtration  
B) Sublimation

## From Confusion to Clarity

— MOLEMIND —

C) Decantation  
D) Magnetic separation

**Q64.** A mixture of two liquids has boiling points that differ considerably. The most suitable method is:

- A) Filtration  
B) Sublimation  
C) Simple distillation  
D) Magnetic separation

**Q65.** Fractional distillation is particularly useful when:

- A) Two solids have different colours  
B) Two liquids have close boiling points  
C) One solid is magnetic  
D) One substance is insoluble in water

**Q66.** Which method is best for separating **oil and water**?

- A) Filtration  
B) Separating funnel  
C) Sublimation  
D) Crystallisation

**Q67.** A student wants to obtain pure water from salt solution. The most appropriate method is:

- A) Filtration  
B) Sedimentation  
C) Distillation  
D) Magnetic separation

**Q68.** During distillation of salt solution, which substance is collected as the distillate?

- A) Salt  
B) Water  
C) Both salt and water  
D) Neither

**Q69.** Why cannot ordinary filtration separate dissolved salt from water?

- A) Salt particles are magnetic  
B) Dissolved salt particles pass through the filter along with water  
C) Water cannot pass through filter paper  
D) Salt evaporates instantly

**Q70.** Crystallisation is preferred over simple evaporation when the objective is to:

- A) Obtain a purer solid
- B) Obtain a liquid
- C) Separate magnetic substances
- D) Separate immiscible liquids

## Experimental & Observation-Based

**Q71.** A student adds sand to water and stirs it. After some time, sand settles at the bottom. This process is:

- Chemistry Builds Better Thinking
- Small Steps Big Dreams :)
- NEET | JEE Foundation
- |   |   |        |
|---|---|--------|
| 1 | H | 1.008  |
| 6 | C | 12.011 |
| 7 | N | 14.007 |
| 8 | O | 15.999 |
- A) Filtration  
B) Sedimentation  
C) Distillation  
D) Sublimation

**Q72.** After sedimentation, the clear upper liquid is carefully poured into another vessel. This process is:

- A) Decantation  
B) Crystallisation  
C) Sublimation  
D) Centrifugation

**Q73.** A muddy water sample is first allowed to stand and then filtered. The purpose of sedimentation before filtration is mainly to:

- A) Increase the amount of dissolved material  
B) Remove larger suspended particles before filtration  
C) Convert the mixture into a compound  
D) Evaporate the water

**Q74.** A student uses chromatography to separate the colours in black ink. The separation occurs because different components:

- A) Have identical solubilities  
B) Move at different rates through the medium  
C) Have identical masses  
D) React chemically with paper

**Q75.** In paper chromatography, a component that travels farther generally has:

- A) Greater attraction to the stationary phase  
B) Greater tendency to move with the solvent  
C) Zero solubility in solvent  
D) Greater density only

**Q76.** A mixture contains two solids: one soluble in water and the other insoluble. Which sequence is appropriate?

- A) Add water → filter → evaporate filtrate  
B) Filter → add water → evaporate residue  
C) Sublime → filter → distil  
D) Distil → filter → evaporate

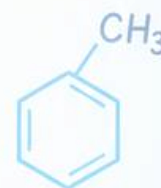
**Q77.** A student heats a mixture containing a substance that sublimates and another substance that does not. The sublimable substance can be collected:

- A) At the bottom of the vessel  
B) On a cooler surface  
C) In the filter paper  
D) In the water layer

**Q78.** Which technique is based mainly on differences in **particle size**?

From Confusion to Clarity

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

Good  
Things  
Take  
Time ♥



A) Filtration  
B) Distillation  
C) Chromatography  
D) Sublimation

**Q79.** Which technique is based mainly on differences in **boiling points**?

A) Filtration  
B) Distillation  
C) Magnetic separation  
D) Sedimentation

**Q80.** Which technique depends on a substance changing directly from solid to vapour?

A) Distillation  
B) Sublimation  
C) Decantation  
D) Filtration

## Scientific Reasoning

**Q81.** A student says, "Air is a compound because it contains oxygen and nitrogen." Which statement correctly identifies the error?

A) Air contains only oxygen  
B) Oxygen and nitrogen are chemically bonded in air  
C) Air is a mixture because its gases are not chemically combined  
D) Air is an element

**Q82.** Which observation most strongly suggests that a sample is a mixture rather than a compound?

A) It has a definite chemical formula  
B) Its components can be separated physically  
C) It has a fixed composition  
D) It has completely new properties

**Q83.** Two colourless liquids are mixed. No gas, precipitate or temperature change is observed, and the mixture can later be separated by distillation. The best conclusion is:

A) They must have formed a compound  
B) They are likely forming a physical mixture  
C) A chemical reaction definitely occurred  
D) Both liquids disappeared

**Q84.** A sample contains particles of different sizes that can be separated by filtration. The sample is most likely:

A) A homogeneous compound  
B) A heterogeneous mixture  
C) An element  
D) A pure substance

**Q85.** Which statement is true for both a compound and a mixture?

Small  
Steps  
Big  
Dreams :)

NEET / JEE  
Foundation

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

s most

## What is the difference between a compound and a mixture?

— MOLEMIND —

NEET

JEE

FOUNDATION

-

5. Soil

Which are mixtures?

- A) 1, 3 and 5
- B) 2 and 4
- C) 1, 2 and 3
- D) 3 and 4

Q92. Which pair is incorrectly matched?

- A) Filtration — insoluble solid + liquid
- B) Distillation — liquids with different boiling points
- C) Sublimation — sublimable solid + non-sublimable solid
- D) Chromatography — magnetic substances

Q93. A student has a mixture of **kerosene, water and dissolved salt**. Which method would be most logical?

- A) Separating funnel → distillation
- B) Filtration → sublimation
- C) Sublimation → filtration
- D) Crystallisation → magnetism

Q94. A pure substance undergoes a change and produces two substances with entirely different properties. The original substance was most likely:

- A) An element
- B) A compound
- C) A mixture
- D) A solution

Q95. Which statement is **most scientifically accurate**?

- A) Every homogeneous substance is a compound
- B) Every heterogeneous substance is a mixture
- C) Every mixture is heterogeneous
- D) Every pure substance contains only one atom

Q96. A student heats a salt solution until all water is removed. The remaining salt is recovered. This demonstrates that:

- A) Salt chemically decomposes in water
- B) Dissolution can be reversed by a physical process
- C) Salt becomes a new compound
- D) Water and salt form a permanent compound

Q97. A mixture of two solids has the following properties:

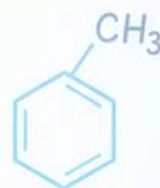
- Solid P is magnetic.
- Solid Q dissolves in water.
- Both are insoluble in kerosene.

Which method is most appropriate?

From Confusion to Clarity

|             |             |             |
|-------------|-------------|-------------|
| 1           | 2           | 3           |
| H<br>1.008  |             |             |
| 4           | 5           | 6           |
|             |             | C<br>12.011 |
| 7           | 8           | 9           |
| N<br>14.007 | O<br>15.999 |             |

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

Good  
Things  
Take  
Time ♥



- A) Add water → filter → magnet  
 B) Magnet → add water → filter  
 C) Add kerosene → filter → evaporate  
 D) Distillation → magnet

**Q98.** A student claims: "If two substances are mixed and no visible change occurs, they must form a mixture." Why is this conclusion not always scientifically sufficient?

- A) A chemical reaction can sometimes occur without an obvious immediate visible change  
 B) All chemical reactions produce bubbles  
 C) Mixtures always change colour  
 D) Compounds can never be colourless

**Q99.** A sample contains only one type of molecule, and each molecule contains atoms of two different elements chemically bonded in a fixed ratio. The sample is:

- A) Element  
 B) Compound  
 C) Homogeneous mixture  
 D) Heterogeneous mixture

**Q100.** A mixture contains **iron filings, ammonium chloride, sand and common salt**. Which sequence gives the most logically complete separation?

- A) Magnet → sublimation → add water → filtration → evaporation  
 B) Sublimation → magnet → add water → evaporation → filtration  
 C) Add water → filtration → magnet → sublimation  
 D) Magnet → filtration → sublimation → distillation

### 🔥 Important Traps from this set

- **Mixture ≠ always heterogeneous** → air, salt solution and brass are homogeneous mixtures.
- **Compound has fixed composition; mixture can have variable composition.**
- **Dissolving ≠ necessarily chemical reaction.**
- **Filtration separates insoluble solids**, not dissolved substances.
- **Simple distillation → appreciably different boiling points.**
- **Fractional distillation → close boiling points.**
- **Sublimation → solid directly → vapour.**
- **Chromatography → different components travel at different rates.**
- **Separation method depends on a physical property** such as magnetism, solubility, particle size or boiling point.

From Confusion to Clarity

— MOLEMIND —

## Atoms, Molecules & Structure of Atom

**Q101.** The smallest particle of an element that retains the chemical properties of that element is:

- A) Molecule
- B) Atom
- C) Ion
- D) Nucleus

**Q102.** Which statement about an atom is correct?

- A) It always exists independently
- B) It is always electrically charged
- C) It is the basic unit of an element
- D) It contains only electrons

**Q103.** Which subatomic particle has a negative charge?

- A) Proton
- B) Neutron
- C) Electron
- D) Nucleus

**Q104.** The mass of an atom is concentrated mainly in its:

- A) Electron cloud
- B) Nucleus
- C) Outer shell
- D) Valence shell

**Q105.** Which particle determines the identity of an element?

- A) Electron
- B) Neutron
- C) Proton
- D) Molecule

**Q106.** An atom has 11 protons. Its atomic number is:

- A) 10
- B) 11
- C) 12
- D) 22

**Q107.** An electrically neutral atom contains 17 protons. Number of electrons is:

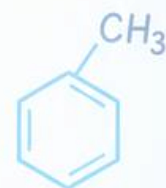
- A) 16
- B) 17
- C) 18
- D) 34

**Q108.** An atom has 12 protons and 12 neutrons. Its mass number is:

NEET | JEE  
Foundation

|   |   |        |
|---|---|--------|
| 1 | H | 1.008  |
| 6 | C | 12.011 |
| 7 | N | 14.007 |
| 8 | O | 15.999 |

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores

Good  
Things  
Take  
Time

From Confusion to Clarity

— MOLEMIND —



A) 12  
B) 24  
C) 6  
D) 0

**Q109.** An atom has mass number 23 and atomic number 11. Number of neutrons is:

A) 11  
B) 12  
C) 23  
D) 34

**Q110.** Isotopes are atoms having:

A) Same mass number, different atomic number  
B) Same atomic number, different mass number  
C) Different atomic number and same neutrons  
D) Same number of electrons only

## Conceptual

**Q11.** Which pair represents isotopes?

A)  $^{12}\text{C}$  and  $^{14}\text{C}$   
B)  $^{14}\text{C}$  and  $^{14}\text{N}$   
C)  $^{23}\text{Na}$  and  $^{24}\text{Mg}$   
D)  $^1\text{H}$  and  $^4\text{He}$

**Q112.** Isotopes of an element generally have similar chemical properties because they have the same:

A) Number of neutrons  
B) Mass number  
C) Electronic configuration  
D) Atomic mass

**Q113.** Which subatomic particle contributes approximately one atomic mass unit?

A) Electron only  
B) Proton and neutron  
C) Electron and proton  
D) Electron and neutron

**Q114.** If an atom loses one electron, it becomes:

A) An anion  
B) A cation  
C) A neutron  
D) An isotope

**Q115.** If an atom gains two electrons, its charge becomes:

A) +2  
B) +1  
C) -1  
D) -2

Small  
Steps  
Big  
Dreams :)

NEET | JEE  
Foundation

Practice  
Learn  
Improve  
Excel

Cc1ccccc1

Better  
Concepts  
Higher  
Scores :)

Good Things Take Time ♥

## From Confusion to Clarity

— MOLEMIND —

NEET

JEE

FOUNDATION

Chemistry  
Builds  
Better  
Thinkers

Small  
Steps  
Big  
Dreams :)

NEET | JEE  
Foundation

**Q116.** The maximum number of electrons that can be accommodated in the first shell is:

- A) 1  
B) 2  
C) 8  
D) 18

**Q117.** The electronic configuration of sodium (atomic number 11) is:

- A) 2, 8, 1  
B) 2, 7, 2  
C) 8, 2, 1  
D) 2, 9

**Q118.** An atom has electronic configuration 2,8,7. It is most likely to:

- A) Lose 7 electrons  
B) Gain 1 electron  
C) Lose 1 electron  
D) Gain 7 electrons

**Q119.** An element with configuration 2,8,2 will most readily form:

- A) 2- ion  
B) 1- ion  
C) 2+ ion  
D) 8+ ion

**Q120.** Which electronic configuration represents a noble gas?

- A) 2,8,1  
B) 2,8,7  
C) 2,8,8  
D) 2,8,6

## Valency & Formulae

**Q121.** The valency of oxygen is generally:

- A) 1  
B) 2  
C) 3  
D) 6

**Q122.** The valency of aluminium is:

- A) 1  
B) 2  
C) 3  
D) 4

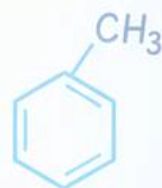
**Q123.** Formula of magnesium chloride is:

ide is:

*From Confusion to Clarity*

— MOLEMIND —

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

Good Things Take Time ♡

NEET

JEE

FOUNDATION

Good Things Take Time ♡

— MOLEMIND —

**Q131.** An ion  $X^{2+}$  has 10 electrons. The atomic number of X is:

- A) 8  
B) 10  
C) 12  
D) 14

**Q132.** An ion  $Y^-$  has 18 electrons. The atomic number of Y is:

- A) 17  
B) 18  
C) 19  
D) 16

**Q133.** An atom has 20 protons, 20 electrons and 20 neutrons. Which is correct?

- A) Atomic number = 20, mass number = 40  
B) Atomic number = 40, mass number = 20  
C) Atomic number = 20, mass number = 20  
D) Atomic number = 40, mass number = 40

**Q134.** Which pair contains particles having the **same number of electrons**?

- A)  $\text{Na}^+$  and Ne  
B)  $\text{Mg}^+$  and Ne  
C)  $\text{Cl}^-$  and  $\text{Ca}^+$   
D)  $\text{Mg}^{2+}$  and  $\text{S}^{2-}$

**Q135.** An element X has configuration 2,8,8,1. Which property is most likely?

- A) It tends to gain one electron  
B) It tends to lose one electron  
C) It is completely unreactive  
D) It tends to gain seven electrons

**Q136.** Which statement about noble gases is most appropriate?

- A) They have incomplete outer shells  
B) They generally have stable outer electronic configurations  
C) They always form +1 ions  
D) They readily react with all metals

**Q137.** An atom has 15 electrons and 16 neutrons. Its mass number is:

- A) 15  
B) 16  
C) 30  
D) 31

**O138.** An element has atomic number 8. Which statement is correct?

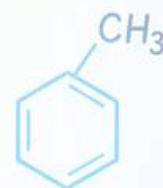
- A) It has 8 neutrons necessarily  
B) It has 8 protons  
C) Its mass number is always 8  
D) It must have 10 electrons

## From Confusion to Clarity

— MOLEMIND —

NEET | JEE  
Foundation

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores :)

Good Things Take Time ♡

NEET

JEE

FOUNDATION

**Q139.** Which species has the greatest number of electrons?

- A)  $\text{Na}^+$   
B)  $\text{Ne}$   
C)  $\text{F}^-$   
D)  $\text{Fe}^{2+}$

**Q140.** If X has valency 3 and Y has valency 2, the likely formula of their compound is:

- A)  $\text{XY}$   
B)  $\text{X}_2\text{Y}_3$   
C)  $\text{X}_3\text{Y}_2$   
D)  $\text{X}_2\text{Y}$

## Reasoning

**Q141.** A student says, "An atom with 8 protons must have a mass number of 8." The statement is:

- A) Always true  
B) Always false  
C) Not necessarily true  
D) True only for ions

**Q142.** Which change does **not** change the identity of an element?

- A) Changing number of protons  
B) Changing number of neutrons  
C) Nuclear transformation changing protons  
D) Changing atomic number

**Q143.** If an atom has equal numbers of protons and electrons, it is:

- A) Always an isotope
- B) Electrically neutral
- C) Always an ion
- D) Negatively charged

**Q144.** Which one has the greatest tendency to form a positive ion?

- A) Element with configuration 2,8,1  
B) Element with configuration 2,8,7  
C) Element with configuration 2,8,8  
D) Element with configuration 2,6

**Q145.** The formula  $\text{MgCl}_2$  indicates that:

- A) One Mg atom combines with one Cl atom  
B) One Mg atom combines with two Cl atoms  
C) Two Mg atoms combine with one Cl atom  
D) Two Mg atoms combine with two Cl atoms

**Q146.** Which pair is correctly matched?

- A)  $\text{Na}^+$  — anion  
B)  $\text{Cl}^-$  — cation

## From Confusion to Clarity

— MOLEMIND —

D)  $O^{2-}$  — cation

A) +3

B) +1

C) -1

D)  $-3$

A)  $\text{Na}^+$

B)  $\text{Mg}^{2+}$

C)  $F^-$

D) All of these

A) Remains sodium

B) Becomes magnesium

C) Becomes an isotope of sodium

D) *Success* Becomes a negative ion

A) X is a neutral chlorine atom

B) X is  $\text{Cl}^-$  with mass number 35

C) X is  $\text{Ar}^-$  with mass number 35

D) X is  $\text{Cl}^+$  with mass number 35

**Q151.** Which observation most strongly indicates a chemical reaction?

### A) Change in size

B) Formation of a new substance

C) Change in state only

### D) Dissolving

**Q152.** Burning magnesium in oxygen produces:

A)  $\text{MgCl}_2$

B) MgO

C)  $\text{MgCO}_3$

D)  $\text{Mg}(\text{OH})_2$

**Q153.** The reaction between an acid and a base producing salt and water is called:

### A) Oxidation

## B) Reduction

### C) Neutralisation

### D) Displacement

**Q154.** Which gas is commonly evolved when an acid reacts with a metal such as zinc?

Small  
Steps  
Big  
Dreams :)

NEET | JEE  
Foundation

Practice  
Learn  
Improve  
Excel

CC1=CC=CC=C1

Better Concepts Higher Scores :)

Good Things Take Time ♥

## From Confusion to Clarity

— MOLEMIND —

Good Things Take Time ♡

D) Distilled water always

— MOLEMIND —

C) Soap solution  
D) Orange juice

**Q170.** An acid reacts with a carbonate to produce:

- A) Hydrogen only  
B) Oxygen only  
C) Carbon dioxide  
D) Nitrogen

## Tricky Conceptual Questions

**Q171.** When dilute HCl is added to sodium carbonate, the gas evolved is:

- A)  $\text{H}_2$   
B)  $\text{O}_2$   
C)  $\text{CO}_2$   
D)  $\text{Cl}_2$

**Q172.** Carbon dioxide produced in the above reaction can be identified because it:

- A) Relights a glowing splint  
B) Turns limewater milky  
C) Burns with a pop sound  
D) Turns blue litmus blue

**Q173.** Antacids are used to relieve excess stomach acidity because they:

- A) Increase acidity
- B) Neutralise excess acid
- C) Produce more acid
- D) Stop digestion completely

**Q174.** Tooth enamel can be damaged when mouth conditions become too acidic. A basic toothpaste can help mainly by:

- A) Increasing acid concentration  
B) Neutralising acids  
C) Producing carbon dioxide  
D) Dissolving enamel faster

**Q175.** Which substance is most suitable for neutralising an acidic agricultural soil?

- A) Quicklime/lime  
B) Vinegar  
C) Lemon juice  
D) Hydrochloric acid

**Q176.** Which substance can be used to neutralise an excessively basic soil?

- A) Suitable organic acidic material  
B) Sodium hydroxide  
C) Lime  
D) Baking soda

## From Confusion to Clarity

— MOLEMIND —



**Q184.** A metal X reacts with dilute HCl but metal Y does not. Which conclusion is reasonable?

- A) Y is necessarily an acid
- B) X is above hydrogen in the relevant reactivity series
- C) X must be less reactive than hydrogen
- D) Y must be a noble gas

**Q185.** Copper does not liberate hydrogen readily from dilute HCl because:

- A) Copper is above hydrogen
- B) Copper is below hydrogen in the reactivity series
- C) HCl is not an acid
- D) Copper is a non-metal

**Q186.** A colourless solution gives a blue colour with blue litmus unchanged but turns phenolphthalein pink. It is most likely:

- A) Basic
- B) Acidic
- C) Neutral
- D) A gas

**Q187.** Which reaction involves **both oxidation and reduction**?

- A) Redox reaction
- B) Neutralisation only
- C) Physical change
- D) Dissolution

**Q188.** In  $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ , CuO is:

- A) Oxidised
- B) Reduced
- C) Neutralised
- D) Sublimed

**Q189.** In the same reaction, hydrogen is:

- A) Oxidised
- B) Reduced
- C) Neither
- D) Converted into an element

**Q190.** Which change would make an acid solution **less acidic**, assuming no other reaction occurs?

- A) Adding more acid
- B) Adding water
- C) Removing water
- D) Increasing acid concentration

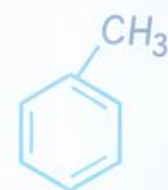
### Grand Challenge

**Q191.** A student performs four experiments:

From Confusion to Clarity

|   |   |   |   |   |             |             |             |   |    |    |    |
|---|---|---|---|---|-------------|-------------|-------------|---|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6           | 7           | 8           | 9 | 10 | 11 | 12 |
|   |   |   |   |   |             | H<br>1.008  |             |   |    |    |    |
|   |   |   |   |   | C<br>12.011 | N<br>14.007 | O<br>15.999 |   |    |    |    |

Practice  
Learn  
Improve  
Excel



Better  
Concepts  
Higher  
Scores

Good  
Things  
Take  
Time

IV.  $\text{CaCO}_3 + \text{HCl} \rightarrow \text{gas}$

## Which experiments provide evidence of chemical reactions?

- A) I only  
B) I and II  
C) I, III and IV  
D) II and IV

**Q192.** A metal reacts with water producing hydrogen. Which property is most relevant in predicting this behaviour?

- A) Colour  
B) Position in reactivity series  
C) Melting point only  
D) Density only

**Q193.** Which combination is correctly matched?

- A) Acid + metal carbonate  $\rightarrow$  H<sub>2</sub>  
B) Acid + metal  $\rightarrow$  H<sub>2</sub>  
C) Acid + base  $\rightarrow$  CO<sub>2</sub>  
D) Base + metal carbonate  $\rightarrow$  O<sub>2</sub>

**Q194.** A solution has pH 5. Another has pH 2. Which is more acidic?

- A) pH 5  
B) pH 2  
C) Both equally  
D) Cannot compare

**Q195.** A student adds a few drops of acid to a basic solution containing phenolphthalein. The pink colour disappears. This indicates:

- A) The solution has become more basic  
B) Acidity has increased enough to neutralise the basic character  
C) Phenolphthalein has become a salt  
D) No reaction occurred

**Q196.** Which statement about neutralisation is most accurate?

- A) It always produces a gas  
B) It generally involves acid + base forming salt and water  
C) It occurs only between strong acids and strong bases  
D) It is always a physical change

**Q197.** A student has four solutions:

- P: pH 1
- Q: pH 6

NEET | JEE  
Foundation

Practice  
Learn  
Improve  
Excel

Cc1ccccc1

four

Better Concepts Higher Scores :)

Good Things Take Time ♡

## From Confusion to Clarity

— MOLEMIND —

NEET

JEE

FOUNDATION

— MOLEMIND —

# ANSWER KEY

| Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans |
|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1  | B   | 11 | A   | 21 | C   | 31 | B   | 41 | B   |
| 2  | C   | 12 | B   | 22 | C   | 32 | D   | 42 | B   |
| 3  | B   | 13 | C   | 23 | B   | 33 | B   | 43 | A   |
| 4  | C   | 14 | B   | 24 | C   | 34 | C   | 44 | A   |
| 5  | C   | 15 | B   | 25 | B   | 35 | A   | 45 | C   |
| 6  | B   | 16 | B   | 26 | A   | 36 | B   | 46 | C   |
| 7  | B   | 17 | C   | 27 | B   | 37 | B   | 47 | B   |
| 8  | B   | 18 | B   | 28 | B   | 38 | B   | 48 | A   |
| 9  | C   | 19 | A   | 29 | A   | 39 | B   | 49 | B   |
| 10 | C   | 20 | C   | 30 | C   | 40 | B   | 50 | C   |

| Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q   | Ans |
|----|-----|----|-----|----|-----|----|-----|-----|-----|
| 51 | B   | 61 | B   | 71 | B   | 81 | C   | 91  | A   |
| 52 | C   | 62 | B   | 72 | A   | 82 | B   | 92  | D   |
| 53 | C   | 63 | B   | 73 | B   | 83 | B   | 93  | A   |
| 54 | D   | 64 | C   | 74 | B   | 84 | B   | 94  | B   |
| 55 | B   | 65 | B   | 75 | B   | 85 | B   | 95  | B   |
| 56 | C   | 66 | B   | 76 | A   | 86 | B   | 96  | B   |
| 57 | A   | 67 | C   | 77 | B   | 87 | B   | 97  | B   |
| 58 | A   | 68 | B   | 78 | A   | 88 | B   | 98  | A   |
| 59 | C   | 69 | B   | 79 | B   | 89 | B   | 99  | B   |
| 60 | C   | 70 | A   | 80 | B   | 90 | C   | 100 | A   |

From Confusion to Clarity

— MOLEMIND —

| Q   | Ans | Q   | Ans | Q   | Ans | Q   | Ans |
|-----|-----|-----|-----|-----|-----|-----|-----|
| 101 | B   | 126 | B   | 151 | B   | 176 | A   |
| 102 | C   | 127 | D   | 152 | B   | 177 | B   |
| 103 | C   | 128 | B   | 153 | C   | 178 | B   |
| 104 | B   | 129 | C   | 154 | C   | 179 | B   |
| 105 | C   | 130 | D   | 155 | B   | 180 | C   |
| 106 | B   | 131 | C   | 156 | B   | 181 | C   |
| 107 | B   | 132 | A   | 157 | B   | 182 | A   |
| 108 | B   | 133 | A   | 158 | A   | 183 | B   |
| 109 | B   | 134 | A   | 159 | B   | 184 | B   |
| 110 | B   | 135 | B   | 160 | A   | 185 | B   |
| 111 | A   | 136 | B   | 161 | B   | 186 | A   |
| 112 | C   | 137 | D   | 162 | B   | 187 | A   |
| 113 | B   | 138 | B   | 163 | A   | 188 | B   |
| 114 | B   | 139 | D   | 164 | C   | 189 | A   |
| 115 | D   | 140 | B   | 165 | B   | 190 | B   |
| 116 | B   | 141 | C   | 166 | A   | 191 | C   |
| 117 | A   | 142 | B   | 167 | B   | 192 | B   |
| 118 | B   | 143 | B   | 168 | C   | 193 | B   |
| 119 | C   | 144 | A   | 169 | C   | 194 | B   |
| 120 | C   | 145 | B   | 170 | C   | 195 | B   |
| 121 | B   | 146 | C   | 171 | C   | 196 | B   |
| 122 | C   | 147 | D   | 172 | B   | 197 | B   |
| 123 | C   | 148 | D   | 173 | B   | 198 | A   |
| 124 | D   | 149 | B   | 174 | B   | 199 | C   |
| 125 | B   | 150 | B   | 175 | A   | 200 | B   |

From Confusion to Clarity

Chemistry  
Builds  
Better  
Thinkers

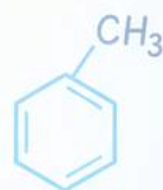
Small  
Steps  
Big  
Dreams :)

NEET | JEE  
Foundation

|                  |                  |                  |
|------------------|------------------|------------------|
|                  |                  | H<br>1.008       |
| 6<br>C<br>12.011 | 7<br>N<br>14.007 | 8<br>O<br>15.999 |



  
Practice  
Learn  
Improve  
Excel  
★



Better  
Concepts  
Higher  
Scores :)

Good  
Things  
Take  
Time ♥



-  Isomerism
-  Electronic Effects
-  Organic Chemistry
-  Concepts  
→ Practice  
→ Success



**MOLEMIND**  
THINK | LEARN | ACHIEVE

Organic  
Chemistry  
is not just a subject...  
it's a way of thinking ♥



From Confusion to Clarity

— MOLEMIND —